Electro mobility acceptance: The influence of political incentives and charging station design

Rafael Kistner, RheinMain University of Applied Sciences

Matthias Kowald, RheinMain University of Applied Sciences

1. Introduction

Greenhouse gas emissions are a major driver of the ongoing climate change, whereby the transport sector is responsible for a substantial proportion of these emissions. E.g. in Germany, the transport sector is the third largest source of emissions, with a share of 18% of the national CO_2-emissions (BMUB 2017). However, increasing numbers of cars and the resulting emissions from combustion engines are also responsible for poor air quality in densely populated areas and cities. Reducing greenhouse gas emissions, to meet the Paris climate agreement, and increase the air quality in cities is the aim of electrification efforts in transportation. For these reasons many countries around the world are trying to promote (private) electric vehicles. Germany for example promised in 2009 to reach a threshold of one million registered electric vehicles by 2020 (Bundesregierung 2009). This aim, however, will be failed. Today only 97.107 electric vehicles are registered in the country, whereby the number did not increase heavily over the last years (ADAC 2019).

According to the progress report of the National Platform for Electromobility (NEP) in Germany, the limited range and the insufficient number of public charging stations are important reasons against purchasing an electric car. The main reason against such an investment, however, is the relatively high price for electric cars in comparison to cars with combustion engines (Anderson et al. 2016). Concerning this later restraint, the German government decided to primarily support customers financially and help cities and regions to build up electric charging infrastructure.

2. Research topic

The city of Rüsselsheim, Germany, will build 650 charging stations with 1.300 charging points in total. From 2020 on, Rüsselsheim with its 63.000 inhabitants, will be the city with the densest charging infrastructure in Europe. The project is part of the "Sofortprogramm Saubere Luft 2017-2020" of the German government, which was launched in 2017. This massive infrastructural roll out transforms the city to a real-world-laboratory and goes hand in hand with accompanying scientific studies, e.g. in terms of planning, organization, and management of the electric grid, consumers´ preferences and acceptance of charging infrastructure and peoples´ cognition of electric mobility.

Our study is part of the accompanying scientific program in Rüsselsheim. It focusses on the acceptance of electric mobility in the population. Our paper will present the methodology of a data collection effort and first results from the field. It focusses on two issues, which are closely related to the acceptance of electric mobility and its diffusion in the German society.

2.1 Incentives and stimuli to promote electric cars

In Germany, municipalities are allowed to set up privileges for electric cars, e.g. in terms of privileged parking, energy costs and access to selected streets. In addition a federal program supports customers in buying an electric car with a financial bonus of up to € 6,000 currently. However, evaluations of the effects of these measurements are rare if available at all. For example, the evaluation of the impact of an increasing financial support is limited to analysing new registrations of electric cars in a given range of time (see de Haan et al. 2018). Applying such a method ignores possible individual and group specific effects, which result from e.g. socio-economic and socio-demographic characteristics and from peoples´ cognitive perception of electric vehicles.

The presented study addresses this effect by analysing peoples´ decisions in a stated adaptation experiment. Participants are presented a number of stimuli, which are varied in accordance to an experimental design. Considering these stimuli, respondents are asked to report an optimal household fleet by providing free choices between all cars and public transport tickets that are available in Germany.

2.2 User-friendly design of charging stations

Currently, there are only few restrictions on charging station operators´ side to designing relevant components of the charging process. For this reason, characteristics like type of authentication, billing, and payment vary between operators and charging stations (Kistner 2019). This heterogeneity results in uncertainty and a negative perception of electro mobility on potential consumers´ side (Klebsch et al. 2017). Employing a choice based conjoint experiment (CBC) aims to compare the utility of different settings of charging station properties and deduce recommendations for optimal designs.

3. Methodology and data collection

To investigate these two research topics, a stated adaptation experiment and a choice-based conjoint experiment are included in a computer assisted personal interview (CAPI) survey. The survey instrument initially records the socio-demographic and socio-economic characteristics of respondents as well as the actual household fleet in terms of cars and public transport abonnements. The annual mobility costs for the reported mobility tools are calculated during the interview and employed as the basis of the stated adaptation experiment.

3.1 Stated adaptation experiment about stimuli for electric vehicle promotion

Stated adaptation experiments belongs to the group of stated response methods (cf. Axhausen and Sammer 2001, Faivre D´Arcier et al. 1998), which aim to confront respondents with hypothetical markets and survey reactions and choices on a predefined set of stimuli presented by the researcher (Janssens et al. 2009).

The stated adaptation approach differs from other stated response methods. It provides a framework for respondents´ choices by presenting stimuli, but it does not present a closed list of choice options as it would be included in e.g. a stated preference study. The research question in a stated adaptation experiment is not how a person would behave in a given situation, as it would be in other stated response methods, but what the respondent would change if the situation was as described (Axhausen and Sammer 2001). Therefore, stated adaptation experiments are characterized by an adaptation of respondents´ behaviour or property, whereby possible reactions and adaptation strategies are open and rather unlimited. For this reason stated adaptation experiments are also referred to as deep and interactive interviews.

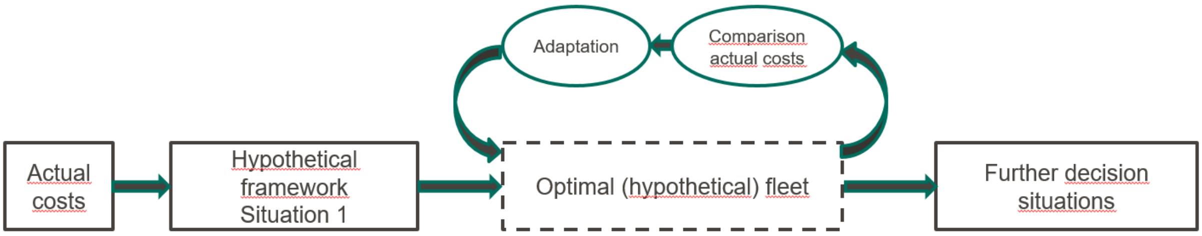
Respondents are asked to answer four choice situations, whereby each situation is characterized by six attributes. These attributes reflect bonus and malus factors that are currently discussed in Germany to promote the selling of electric cars. The attributes vary in a realistic way by including enough range to potentially result in behavioural changes. An overview on the experimental design is provided in Table 1.

Table 1: Stated Adaptation experiment – overview on experimental design

Variable	*parameter value 1*	*parameter value 2*	*parameter value 3*
petrol costs (€/l)	1,50	3,00	4,50
electricity price (€/100 km)	0,00	3,50	7,00
CO-2 surcharge (per liter of petrol)	free	+5 cent	+15 cent
environmental / purchase bonus (€)	2.000	6.000	10.000
tax for electric cars	free	50% of ICE vehicles	like ICE vehicles
public transport prices relative to today	free	50% of todays price	as today

The stated adaptation approach allows an analysis of the effect of political bonus and malus factors. In reaction to the provided stimuli respondents are asked to optimize their household fleet. To avoid unrealistic choices and household fleet compositions, the annual mobility costs for the fictive fleet are calculated in accordance to the shown stimuli and compared to the costs of the actual fleet. Respondents are asked to consider changes in mobility costs and judge if their household can manage possible gaps between actual and fictive fleet. If this evaluation leads to a negative result, respondents are asked to further optimize their decision until a fleet composition is found that fits the stimuli from the choice situation in an optimal way and is realistic with respect to the respondent´s financial restrictions. The iterative approach of the adaptation process is illustrated in Figure 1.

Figure 1: Scheme of the adaptation process in a choice situation



Statistical analysis of the Stated Adaptation Experiment will be done in form of regression models on a binary, dependent variable that indicates whether and under which conditions a person decides to include an electric car in the household fleet. In addition, multiple regression models on a continuous dependent variable intend to provide information about the main effects in relation to change in gasoline consumption and CO_2 emissions of the household fleet. Analysis consider respondents´ socio-demographic and socio-economic background and can thus be performed for different target groups. Finally, structural equation models identify the causality between manifest effects from the decision situation and respondents´ cognitive perception of electric cars (see section 3.3 of this abstract).

3.2 Conjoint Analysis on design options for charging stations

Choice based conjoint experiments (CBC) allow an analysis of different product settings and designs (Backhaus et al. 2008, Gustafsson et al. 2003). In our study, such an experiment is employed to analyse different design options for charging stations. For this purpose, different forms of relevant design properties are considered in the experimental design. Respondents are asked to choose between two different configurations of a charging station, whereby each participant is asked to make a decision in twelve choice situations. Modelling respondents´ choices show the preferred characteristics and the preferred combinations of these characteristics, e.g. if a payment by credit card has a high or low utility, or if a metering system relying on loading time or energy consumption is preferred. Table 2 shows summarizes the experimental design for the choice based conjoint experiment.

Table 2: Choice based conjoint experiment – Overview on experimental design

Variable	*parameter value 1*	*parameter value 2*	*parameter value 3*	*parameter value 4*
Authentification	Plug & Charge	RFID	App	
Payment	Web-based	cards based	debit procedure	
Billing	according to the amount of electricity	by time	lump sum	Flatrate
Share of electricity from regenerative energy sources	0%	50%	100%	

3.3 Cognitive perception of electromobility

In addition to the two experiments, participants are asked to provide information about their cognitive perception of electro mobility. This part of the questionnaire employs constructs from socio-psychological theories like the theory of planned behavior (Ajzen, 1991) and the unified theory of acceptance and use of technology (Venkatesh, 2003), which focus on intrinsic motivations and are often used when analyzing the intention to use a certain transport mean (Bamberg et al. 2000; Heath and Gifford 2002; Haustein and Hunecke 2007). The questionnaire includes items on attitudes, norms, and self-efficacy. Related questions are if people consider an electric car as useful or harmful or if they think they can handle the charging process easily or not. Personal norms describe behavioral patterns which respondents consider as being expected from them by their social environment and, especially, by social core contacts like family members. Items on self-efficacy describe peoples´ believe to be able to successfully perform a behavior with the individually available resources. For example, if they believe to find an available charging point, when the battery of their car is low. All items for the socio-psychological constructs were identified in a qualitative study employing elicitation techniques and tested in a quantitative pretest.

4. Contribution to the paper

In September 2020, at the time the hEART conference will take place, the survey will be completed and presumably 400 respondents will be interviewed. The paper will present the survey instrument and fieldwork. In addition, first models and related results from the two choice experiments and on the socio-psychological items will be presented. These first analysis will show how the bonus and malus factors help to increase the selling of electric cars and which attributes of charging stations are preferred by potential customers. The paper closes with an outlook on further research.

Literature

ADAC (Ed.) (2019): Elektroautos 2019: Das sind die Neuen. [online] https://www.adac.de/rund-ums-fahrzeug/e-mobilitaet/kaufen/neue-elektroautos/ (last consulted on 26.02.2019)

Anderson, J. et al. (2016): LADEN2020. Schlussbericht. Konzept zum Aufbau einer bedarfsgerechten Ladeinfrastruktur in Deutschland von heute bis 2020. Deutsches Zentrum für Luft- und Raumfahrt e.V. Karlsruher Institut für Technologie. [online] https://elib.dlr.de/111054/2/LADEN2020_Schlussbericht.pdf (last consulted on 17.02.2019)

Axhausen K. and G. Sammer (2001): Stated Responses: Überblick, Grenzen, Möglichkeiten. Internationales Verkehrswesen, 34 (3-4).

Backhaus K., Erichson B., Plinke W. and R. Weiber (2008): Multivariate Analysemethoden. Eine anwendungsorientierte Einführung. Springer Verlag. Berlin. Heidelberg.

Bamberg, S.; Gumbl, H. and P. Schmidt (2000): Rational Choice und theoriegeleitete Evaluationsforschung: Am Beispiel der "Verhaltenswirksamkeit verkehrspolitischer Maßnahmen". Springer. Wiesbaden.

Bundesministerium für Umwelt, Naturschutz, Bau und Reaktorsicherheit (Ed.) (2017): Klimaschutz in Zahlen: Der Sektor Verkehr. [online] https://www.bmu.de/fileadmin/Daten_BMU/Download_PDF/Klimaschutz/klimaschutz_in_zahlen_verkehr_bf.pdf (last consulted on 17.02.2019)

Bundesregierung (Ed.) (2009): Nationaler Entwicklungsplan Elektromobilität der Bundesregierung. [online] https://www.erneuerbar-mobil.de/sites/default/files/2016-08/nep_09_bmu_bf.pdf (last consulted on 19.02.2019)

De Haan P., Bianchetti R., Rossner S. and H. Frantz (2018): Szenarien der Elektromobilität in Deutschland. EBP-Hintergrundbericht. Zollikon/Berlin.

Faivre D´Arcier B., Andan O. and C. Raux (1998): Stated adaptation surveys and choice process: Some methodological issues. Transportation, 25 (2).

Gustafsson A., Herrmann A. and F. Huber (2003): Conjoint Measurement. Methods and Applications. Springer Verlag. Berlin. Heidelberg.

Haustein, S. and M. Hunecke (2007): Reduced Use of Environmentally Friendly Modes of Transportation Caused by Perceived Mobility Necessities: An Extension of the Theory of Planned Behavior. Journal of Applied Social Psychology. 37.8.

Heath, Y. and R. Gifford (2002): Extending the Theory of Planned Behavior: Predicting the Use of Public Transportation. Journal of Applied Social Psychology.32.2154-2189.

Janssens D., Cools M., Moons E., Wets G., Arentze T. and H. Timmermans (2009): Road Pricing as an Impetus for Environment-Friendly Travel Behavior. In: Transportation Research Record: Journal of the Transportation Research Board, No. 2115, Transportation Research Board of the National Academies, Washington, D.C., 2009, 50–59.

Kistner R. (2019): Ladekonzepte, Ladesäulengestaltung, Meteringverfahren und Abrechnungssysteme für elektrische Antriebe im MIV – eine Übersicht. Masterthesis. Hochschule RheinMain.

Klebsch W. et al. (2017): Ad-hoc-Laden und spontanes Bezahlen. Wie sich „punktuelles Auf-laden“ umsetzen lässt. VDE Verband der Elektrotechnik, Elektronik und Informationstechnik e.V. Frankfurt /M. Online im WWW unter URL: https://www.ebg-com-pleo.de/fileadmin/content/compleo/downloads/OVAL-Studie_Download.pdf (last consulted on 01.03.2019)